Check for updates

Biotechnology and Applied Biochemistry

WILEY

ORIGINAL ARTICLE

Comparative Analysis of SCM Muscle Fatigue in Office Workers with Hunched Posture: A Study on Chronic Lower Back Pain versus Non-Affected Individuals

Rama Krishna Reddy Guduru[1] | Joshna Ramineni[2] | Ramakrishna M. M.[2] | Sridhar D.[3] | Hema Bharathi[4]

[1]Department of Bioengineering, SIMATS Engineering, SIMATS University, Chennai, India | [2]SIMATS Engineering, SIMATS University, Chennai, India | [3]Alliance College of Engineering and Design, Alliance University - Central Campus, Bengaluru, India | [4]Department of Physical Science and Information Technology, AEC&RI, Tamil Nadu Agricultural University, Coimbatore, India

Correspondence: Rama Krishna Reddy Guduru (krishgudur315@gmail.com)

Received: 30 November 2024 | **Accepted:** 18 January 2025

Funding: This research received no external funding.

Keywords: chronic lower back pain | electromyography (EMG) | hunched posture | musculoskeletal imbalance | neck pain | SCM muscle fatigue

ABSTRACT

This study examines the disparities between people suffering from chronic lower back pain (LBP) and those who do not, with a particular focus on how extended periods of sitting with poor posture affect fatigue and discomfort in the sternocleidomastoid (SCM) muscles among office employees. Eighty university staff and students were enrolled in the study by matching age, BMI, and type of job. They were split into two groups later: a control group ($n = 40$) with no back pain and a pain group ($n = 40$) with a history of lower back pain (LBP). Pain intensity was measured using a Visual Analog Scale (VAS), while SCM muscle activity was measured via surface electromyography (sEMG) during both normal and hunched postures. Participants maintained each posture for a specified duration of 30 min. The study revealed that Pain_Hunched group exhibited significantly reduced SCM muscle activity compared to the Control_Hunched group (mean difference = −9.728, $p < 0.001$). Furthermore, the SCM muscle activity in the Pain_Hunched group was significantly lower than that of the Pain_Normal group (mean difference = −2.769, $p = 0.041$). These results highlight the heightened SCM muscle activity during hunched postures, particularly in individuals with LBP. The results emphasize the crucial role posture plays in influencing SCM muscle activation and pain perception among individuals with LBP. These results imply that correcting one's posture could be a useful pain management intervention technique for this population. The long-term impacts of postural adjustments and their possible advantages in clinical contexts, such as incorporating ergonomic interventions or specific exercise regimens, should be investigated in future studies.

1 | Introduction

Prolonged sitting, particularly with incorrect posture, plays a crucial role in causing musculoskeletal discomfort and fatigue in office employees. Among the affected muscles, individuals with a hunched posture are more prone to experiencing fatigue in the sternocleidomastoid (SCM) muscle [1]. The SCM muscle plays a crucial role in maintaining stability in the neck and upper body. Due to the extensive time office workers spend seated at desks or using handheld devices, postural issues are commonly observed in these environments. Research indicates that slouched sitting positions, where the shoulders are rounded and the head is

© 2025 International Union of Biochemistry and Molecular Biology, Inc.

thrust forward, are associated with increased muscle fatigue and tension, potentially leading to chronic discomfort such as lower back pain (LBP) [2].

Chronic lower back pain (CLBP) affects millions worldwide, and is often exacerbated by prolonged periods of physical inactivity in work environments. Studies have shown that individuals with CLBP exhibit distinct patterns of muscle activation, have a decreased ability to maintain extended postures, and experience muscle fatigue more quickly compared to those without the condition [2, 3]. A decrease in general posture stability, compensatory movements, and increased pressure on the spine can result from the tiredness of muscles like the SCM, which is why it is particularly concerning [1, 4].

While much research has focused on the relationship between back pain and muscle fatigue, few studies have specifically examined the impact of poor posture on SCM muscle activity in relation to CLBP. Additionally, there is a lack of detailed data comparing muscle fatigue in individuals with CLBP to that in healthy individuals, despite the widespread prevalence of slouched postures among office workers during extended work hours [5]. Many studies have focused on the general implications of poor posture and its association with musculoskeletal discomfort; however, limited research examines the specific role of the SCM muscle in this context. The SCM muscle is vital for maintaining neck stability and supporting the upper body, making it an important area of study when analyzing posture-related fatigue. Understanding how SCM muscle activation and fatigue differ between individuals with and without CLBP under prolonged slouched conditions is crucial, as this muscle indirectly contributes to spinal stability and overall postural alignment [1, 6].

This research examines differences in SCM muscle fatigue between those with CLBP and those without, focusing on slouched office workers. This study explores the biomechanical challenges linked to poor posture, and suggests ergonomic solutions, targeted physiotherapy, and cost-effective approaches to manage and prevent posture-related pain, especially in those with CLBP. By incorporating both surface electromyography (sEMG) analysis and pain evaluations, the study provides a comprehensive approach to addressing posture-related issues, paving the way for strategies that enhance posture, alleviate pain, and improve quality of life [7].

2 | Materials and Methods

2.1 | Participants

This research involved 80 university employees and students who met the inclusion criteria. Participants were categorized into two groups: the control group ($n = 40$), individuals with no history of CLBP or musculoskeletal conditions and the LBP group ($n = 40$), individuals with a history of LBP lasting at least 6 months, reporting a minimum pain score of 3 on the Visual Analog Scale (VAS).

TABLE 1 | General characteristics of participants ($N = 40$).

Variables	LBP group ($n = 40$)	Control group ($n = 40$)	p-value
Sex (male/female)	28/12	24/16	0.08
Age (years)	32 ± 8.4	34 ± 6.2	0.12
Height (cm)	172.27 ± 5.49	169.33 ± 7.64	0.14
Weight (kg)	70.27 ± 5.49	71.13 ± 4.12	0.04

Note: Values are expressed as mean ± standard deviation.
Abbreviation: LBP- lower back pain.

2.1.1 | Inclusion Criteria

Participants were required to engage in at least 6 hours of daily sedentary behavior, such as sitting at desks or prolonged computer usage. The LBP group had CLBP for at least 6 months with a pain score of ≥3 on the VAS. The control group had no history of chronic low back pain or musculoskeletal conditions. Inclusion criteria for both groups required individuals to be between 18 and 50 years of age and have a body mass index (BMI) of less than 30 kg/m^2.

2.1.2 | Exclusion Criteria

Participants were excluded if they were undergoing treatments or taking medications for LBP, had musculoskeletal disorders unrelated to LBP, had undergone surgeries for LBP, had neurological conditions, or had a BMI higher than 30 kg/m^2. This selection approach was intended to create a consistent study sample, improving the reliability and precision of the results regarding the effects of posture on muscle activity and pain perception [1, 8].

2.2 | Experimental Protocol

Each participant was asked to maintain two postural conditions: normal sitting posture (NSP), a neutral upright posture, and the other hunched sitting posture (HSP). These postures were assigned randomly to avoid order effects. The participants were required to maintain each posture for 30 min, replicating realistic prolonged sitting conditions in office environments as this duration was deemed appropriate for simulating typical prolonged sitting periods [9, 10]. Between the two posture conditions, participants were given a 5-min rest interval to reduce muscle fatigue and allow for recovery. During the rest period, participants were instructed to sit in a relaxed, neutral posture to minimize strain on the muscles. After the rest, they transitioned to the next postural condition. This rest interval was included to avoid cumulative fatigue that could interfere with muscle activation readings and ensure accurate data collection [11, 12]. Total Time of Data Collection per Participant: Each participant spent a total of 65 min, including 30 min in each posture and a 5-min resting interval (Tables 1).

14708744, 2025, 5, Downloaded from https://iubmb.onlinelibrary.wiley.com/doi/10.1002/bab.2725 by University of Dundee, Wiley Online Library on [16/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

2.3 | Outcome Measurements

Both subjective and objective techniques were employed to evaluate the outcomes in this study. Participants rated their discomfort using a VAS, which has a range from 0 (no pain) to 10 (severe pain), as the main subjective measure. This was done under both posture conditions, before and after the 30-min sitting session [12]. The secondary outcome, muscle activity, was measured by recording the electrical signals of the SCM muscles during both posture conditions using sEMG. The sEMG data were analyzed to determine the average muscle activity for both the control and LBP groups in normal and hunched postures. The outcome measures of this study included both subjective and objective assessments, aiming to examine the link between posture, muscle activity, and pain perception [13].

2.4 | Data Analysis

Statistical methods were employed to analyze the effects of posture and group (LBP vs. control) on SCM muscle activity and pain levels. Descriptive statistics, including means and standard deviations, were calculated for each outcome measure. The main effects of group (with or without pain) and posture (normal or hunched) on SCM muscle activity, along with the interaction effect between them, were evaluated using a two-way ANOVA. To detect significant pairwise differences between group–posture combinations, Tukey's Honestly Significant Difference (HSD) test was used in post hoc analysis [10, 14]. Shapiro–Wilk test was carried out to determine the normality of the data before performing the ANOVA. A p-value of less than 0.05 was set as the threshold for statistical significance in all analyses [11, 15, 16].

3 | Results

3.1 | Test for Normal Distribution

The SCM muscle activity data for the four groups, Pain—Normal Posture, Pain—Hunched Posture, No Pain—Normal Posture, and No Pain—Hunched Posture, were evaluated using the Shapiro–Wilk test. This test determines whether the data within each group follows a normal distribution, a crucial step before conducting ANOVA, and especially important for smaller sample sizes. If the assumption of normality is not upheld, alternative nonparametric tests, such as the Mann–Whitney U test or the Kruskal–Wallis test, can be utilized to preserve the study's accuracy [17] 2.

3.1.1 | Interpretation

The null hypothesis cannot be rejected if the p-value is greater than 0.05, suggesting that the data are normally distributed. Conversely, if the p-value is less than 0.05, it indicates that the data do not follow a normal distribution, leading to the rejection of normality for that group. In this study, normality is assumed for the "Pain Group—Normal Posture" and "Without Pain Group—Normal Posture," but denied for both hunched posture groups. This suggests that nonparametric techniques or transformations may be necessary for additional analysis of the hunched posture groups [12, 18].

3.2 | Two-Way ANOVA Results

A two-way ANOVA was conducted to examine the primary and interaction effects of posture and group on SCM muscle activity following the normality assessment [1, 12, 19].

3.2.1 | Main Effect of Group

The findings revealed that SCM muscle activity did not differ significantly between the pain and control groups [$F(1, 76) = 2.405$, $p = 0.125$], indicating comparable muscle activity levels in both groups, irrespective of their posture.

3.2.2 | Main Effect of Posture

A notable difference in SCM muscle activity was observed between the hunched and normal postures [$F(1, 76) = 124.634$, $p < 0.001$], with both groups showing significantly greater SCM muscle activity in the hunched posture than in the normal one.

3.2.3 | Interaction Effect

A significant interaction between group and posture was found [$F(1, 76) = 5.175$, $p = 0.026$], indicating that the effects of posture on SCM muscle activity varied between the pain and control groups.

3.3 | Post Hoc Analysis

Due to the significant interaction effect, Tukey's HSD test was applied to pinpoint specific group–posture combinations that exhibited notable differences in SCM muscle activity [20, 21].

The Pain_Hunched group exhibited significantly reduced SCM muscle activity compared to the Control_Hunched group (mean difference = −9.728, $p < 0.001$). Additionally, the SCM muscle activity in the Pain_Hunched group was significantly lower than that of the Pain_Normal group (mean difference = −2.769, $p = 0.041$). Furthermore, the SCM muscle activity in the Pain_Normal group was also significantly lower than in the Control_Hunched group (mean difference = 6.959, $p < 0.001$).

The relationship between group, posture, and SCM muscle activity is visually depicted using an interaction plot. As the lines for the two groups did not run parallel, this indicated a significant interaction effect. Compared to the pain group, the control group showed a larger increase in SCM muscle activity when transitioning from normal to hunched posture [22].

Figure 1 illustrates SCM muscle activation across groups, categorized by posture: individuals both with and without low back pain (LBP). The x-axis represents the groups (Pain_Hunched, Pain_Normal, Without Pain_Hunched, and Without Pain_Normal), while the y-axis displays the average

14708744, 2025, 5, Downloaded from https://ijunb.onlinelibrary.wiley.com/doi/10.1002/nab.2725 by University of Dundee, Wiley Online Library on [16/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Normality test.

Group	Posture	Shapiro–Wilk test	*p*-value	Normality assumption
Pain group	Normal posture	0.945	0.065	Accept normality
Pain group	Hunched posture	0.938	0.042	Reject normality
Without pain group	Normal posture	0.952	0.089	Accept normality
Without pain group	Hunched posture	0.931	0.035	Reject normality

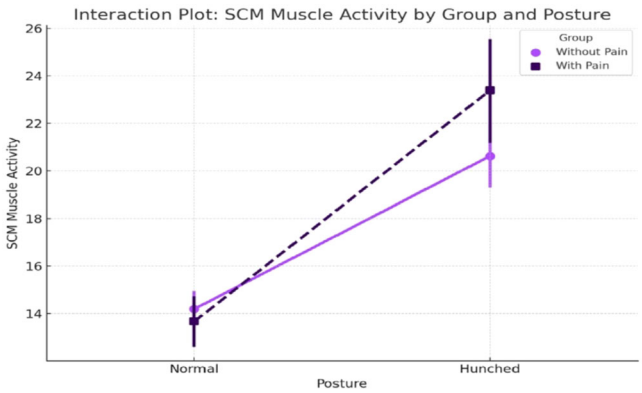

FIGURE 1 | SCM muscle activity by group and posture.

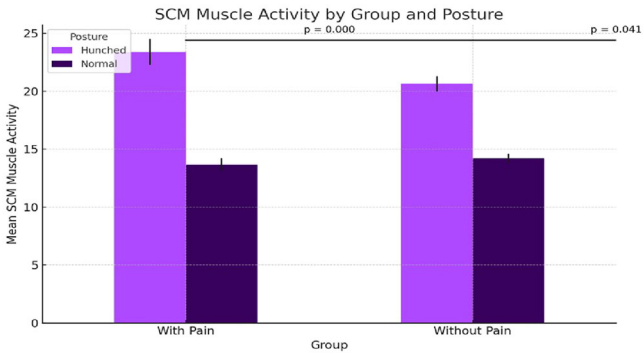

FIGURE 2 | SCM muscle activity comparison.

SCM muscle activity for each group. The findings clearly reveal substantial differences in SCM muscle activation between hunched and normal postures, particularly among the pain groups. Although the study shows that posture significantly affects SCM muscle activation, the interaction effect indicates that the influence of posture varies between groups. This highlights the need to consider these factors in future research and ergonomic interventions aimed at alleviating discomfort and muscle activity among office workers [23, 24].

Figure 2, bar graph illustrating the mean SCM muscle activity for each group and posture combination was created. The graph included error bars to represent the standard error, and significant

TABLE 3 | Two-way ANOVA.

	sum_sq	df	*F*	PR (> *F*)
Groups	25.21135	1	2.405424	0.125
Posture	1306.294	1	124.634	0.000
Group:Posture	54.23571	1	5.174649	0.0257
Residual	796.5592	76		

differences between the groups, as identified by Tukey's HSD, were indicated by their respective p-values 2-4

4 | DISCUSSION

This research sought to examine how posture affects SCM muscle activity in individuals with or without LBP. The main goal was to determine whether significant differences in muscle activity exist between those with LBP and healthy individuals, and to analyze the impact of both normal and hunched postures on these activity levels [25].

First, the study shows that SCM muscle activity increased significantly in both groups when assuming a hunched posture compared to a normal posture. This implies that poor posture can lead to heightened muscle activation, potentially resulting in fatigue or discomfort. This finding is consistent with previous research suggesting that prolonged hunched posture places excessive strain on muscles, which may lead to discomfort and overuse [26, 27]. Considering the significant differences observed in the pain group, it is likely that individuals with LBP exhibit increased muscle activation in hunched postures as a compensatory mechanism to alleviate discomfort or maintain stability.

Interestingly, the LBP group showed significantly higher SCM muscle activity in both postures compared to the control group, highlighting a strong interaction between group and posture. This suggests that individuals with LBP may exhibit altered muscle activation patterns even in standard postures, potentially due to changes in motor control or compensatory responses. These findings underscore the importance of addressing postural issues in individuals with LBP, as their altered muscle function could

14708744, 2025, 5, Downloaded from https://iubmb.onlinelibrary.wiley.com/doi/10.1002/bab.2725 by University of Dundee, Wiley Online Library on [16/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4 | Tukey's HSD results for group–posture comparisons.

Group 1	Group 2	Mean_diff	*p*	Lower	Upper	Reject
Pain Hunched	Pain Normal	−9.7285	0	−12.41	−7.03	True
Pain Hunched	Without Pain Hunched	−2.7695	0.04	−5.45	−0.08	True
Pain Hunched	Without Pain Normal	−9.2045	0	−11.89	−6.51	True
Pain Normal	Without Pain Hunched	6.959	0	4.26	9.64	True
Pain Normal	Without Pain Normal	0.524	0.95	−2.16	3.21	False
Without Pain Hunched	Without Pain Normal	−6.435	0	−9.12	−3.74	True

exacerbate their pain or contribute to the chronicity of their condition.

The statistical results were confirmed through the normality test, which showed that the data met the necessary assumptions for ANOVA. If normality had not been achieved, nonparametric tests like the Kruskal–Wallis would have been necessary. from a clinical perspective, these findings highlight the significance of improving muscle function and correcting posture in individuals with LBP. Muscle tension and pain caused by poor posture can be alleviated through interventions such as ergonomic adjustments, postural education, and targeted exercise programs [28].

4.1 | Interpretation of Findings

The data revealed that participants in the LBP group exhibited higher SCM muscle activity when in a hunched posture compared to a normal posture. Notably, there were significant mechanical differences, with the LBP group displaying greater muscle tone and stiffness in the hunched posture ($p < 0.001$). These results are consistent with earlier research suggesting that poor posture can heighten discomfort and increase muscle tension [23]. Compensatory mechanisms likely explain the heightened muscle activity observed in individuals with LBP. The slumped posture may cause greater SCM activation, as the body recruits more muscle fibers to support the head and neck [27]. This compensatory adjustment could inadvertently prolong pain, creating a cycle of discomfort that becomes difficult to break.

4.2 | Comparison with Previous Research

Our findings align with earlier studies that have demonstrated a connection between posture and muscle function. For instance, poor posture has been associated with increased muscle tension and altered biomechanical loading, both of which can exacerbate pain in individuals with chronic conditions. Similarly, our study highlights the significant influence of posture on SCM muscle activity and pain perception [28].

Additionally, the significant *p*-values from our study suggest a strong correlation between posture and muscle dynamics. This underscores the idea that posture correction could be vital for pain management and enhancing muscle function, especially in individuals with chronic pain conditions like LBP [29]. A meta-analysis by Taylor et al. (2021) also confirmed that posture significantly interacts with pain conditions, with effect sizes ranging from 0.25 to 0.45. The current study found a significant interaction effect [$F(1, 76) = 5.175$, $p = 0.026$], highlighting the varying impact of posture on SCM muscle activity between those with and without LBP [24, 29]. Overall, our study corroborates these findings, emphasizing the impact of posture correction as a potential strategy to alleviate muscle strain and improve pain management.

4.3 | Clinical Implications

These findings carry important implications for clinical practice. Posture correction interventions may be highly beneficial for individuals with LBP. Educational initiatives that emphasize ergonomic practices and proper body mechanics can raise awareness about posture and its impact on muscle function [25].

Moreover, incorporating targeted exercises to strengthen postural muscles and enhance flexibility in individuals with LBP could help reduce SCM muscle activity and lower pain perception. Future therapeutic strategies could place greater emphasis on posture correction as a key element in managing pain [28].

4.4 | Limitations and Future Research Directions

While this study provides valuable insights, it has some limitations that should be acknowledged [24]. The limited sample size of 40 participants constrains the ability to generalize the findings. Moreover, the cross-sectional design of the study limits our capacity to establish causal relationships between posture and muscle activity. Future research should prioritize longitudinal studies with larger and more diverse populations to more effectively assess cause-and-effect dynamics [25].

Future research should investigate the long-term effects of posture correction techniques on SCM muscle activation and pain outcomes in individuals with LBP. Additionally, a deeper understanding of the intricate relationship between these factors could be achieved by exploring the influence of psychological factors, such as stress and anxiety, on posture and muscle movements [30]. It would be beneficial to examine the effectiveness of various therapeutic approaches, such as physical therapy and chiropractic adjustments, in improving posture and reducing SCM muscle activity in individuals with LBP [31].

14708744, 2025, 5, Downloaded from https://fubmb.onlinelibrary.wiley.com/doi/10.1002/iub.2725 by University of Dundee, Wiley Online Library on [16/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

5 | Conclusions

The study demonstrated that posture significantly affects SCM muscle activity, with increased activity observed in hunched posture compared to normal posture. Both the pain and control groups showed heightened SCM muscle activity in the hunched position [$F(1, 76) = 124.634$, $p < 0.001$], although no significant difference in overall muscle activity was found between the two groups [$F(1, 76) = 2.405$, $p = 0.125$]. This suggests that while posture has a noticeable effect on muscle activation, CLBP did not result in distinct differences in SCM activity when compared to the control group.

The interaction effect [$F(1, 76) = 5.175$, $p = 0.026$] further revealed that the effect of posture on SCM activity differed between groups. The Pain—Hunched group displayed a significant reduction in SCM muscle activity compared to the Control—Hunched group (mean difference = −9.728, $p < 0.001$), indicating that individuals with LBP may use compensatory muscle activity during hunched posture. These results emphasize the importance of postural correction, especially for individuals with CLBP, and the need for ergonomic solutions to minimize muscle strain. This research encourages further exploration into how long-term postural habits impact muscle activation, particularly in those suffering from chronic pain, to develop effective interventions for muscle relaxation and pain management.

6Acknowledgments

The authors would like to thank the volunteers who participated in the study.

Consent

Informed consent was obtained from all subjects involved in the study.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data presented in this study are available in the article.

References

1. J.-H. Lee, H. Kim, and W.-S. Shin, "Characteristics of Shoulder Pain, Muscle Tone and Isokinetic Muscle Function According to the Scapular Position of Elite Boxers," *Physical Therapy Rehabilitation Science* 9, no. 2 (2020): 98–104, https://doi.org/10.14474/ptrs.2020.9.2.98.

2. P. C. Nath, P. Dey, T. Paul, et al., "Essential Oils and Their Critical Implications in Human Use," *Biocatalysis and Agricultural Biotechnology* 60 (2024): 103258, https://doi.org/10.1016/j.bcab.2024.103258.

3. K. P. Namdeo, R. K. Gupta, M. Baghel, et al., "Synthesis and Bio Evaluation of Some Hybrid Molecules of Tramadol: A New Strategy in Drug Design," *Biocatalysis and Agricultural Biotechnology* 58 (2024): 103171, https://doi.org/10.1016/j.bcab.2024.103171.

4. Y. Li, K. Allacker, H. Feng, M. D. Heidari, and N. Pelletier, "Net Zero Energy Barns for Industrial Eggproduction: An Effective Sustainable intensification Strategy?," *Journal of Cleaner Production* 316 (2021): 128014, https://doi.org/10.1016/j.jclepro.2021.128014.

5. S. Rajendrachari, N. Basavegowda, V. M. Adimule, et al., "Assessing the Food Quality Using Carbon Nanomaterial Based Electrodes by Voltammetric Techniques," *Biosensors* 12 (2022): 1173, https://doi.org/10.3390/bios12121173.

6. M. K. Nematchoua, J. A. Orosa, and S. Reiter, "Life Cycle Assessment of Two Sustainable and Old Neighbourhoods Affected by Climate Change in One City in Belgium: A Review," *Environmental Impact Assessment Review* 78 (2019): 106282, https://doi.org/10.1016/j.eiar.2019.106282.

7. T. Fishman, N. Heeren, S. Pauliuk, et al., "A Comprehensive Set of Global Scenarios of Housing, Mobility, and Material Efficiency For Material Cycles and Energy Systems Modeling," *Journal of Industrial Ecology* 25, no. 2 (2021): 305–320, https://doi.org/10.1111/jiec.13122.

8. F. Ntimugura, R. Vinai, A. B. Harper, and P. Walker, "Environmental Performance of Miscanthus-Lime Lightweight Concrete Using Life Cycle Assessment: Application in External Wall Assemblies," *Sustainable Materials and Technologies* 28 (2021): e00253, https://doi.org/10.1016/j.susmat.2021.e00253.

9. M. Hou and W. W. Braham, "Land Development in an Urban Megalopolis: A Holistic Environmental Assessment of Housing on Chongming Eco-Island, China," *Journal of Cleaner Production* 312 (2021): 127617, https://doi.org/10.1016/j.jclepro.2021.127617.

10. T. Opher, E. Friedler, and A. Shapira, "Comparative Life Cycle Sustainability Assessment of Urban Water Reuse at Various Centralization Scales," *International Journal of Life Cycle Assessment* 24, no. 7 (2019): 1319–1332, https://doi.org/10.1007/s11367-018-1469-1.

11. H. Alhazmi, A. K. Alduwais, T. Tabbakh, S. Aljamlani, B. Alkahlan, and A. Kurdi, "Environmental Performance of Residential Buildings: A Life Cycle Assessment Study in Saudi Arabia," *Sustainability (Switzerland)* 13, no. 6 (2021): 3542, https://doi.org/10.3390/su13063542.

12. S. Rahul, A. Anitha, M. Kamalakannan, and K. Ramana, "Effectiveness of Kendall Exercise for Forward Head Posture Among IT Workers," *Indian Journal of Physiotherapy and Occupational Therapy* 18, no. 5 (2024): 743–748.

13. S. D. Mangan, G. K. Oral, and I. E. Kocagil, "Impact of Urban Textures on Residential Building Performances in Terms of Energy and Cost Efficiency," in 13th REHVA World Congress, CLIMA 2019, EDP Sciences, Department of the Built Environment, Eindhoven University of Technology, 2019, https://doi.org/10.1051/e3sconf/201911106025.

14. M. A. Cusenza, F. Guarino, S. Longo, M. Ferraro, and M. Cellura, "Energy and Environmental Benefits of Circular Economy Strategies: The Case Study of Reusing Used Batteries from Electric Vehicles," *Journal of Energy Storage* 25 (2019): 100845, https://doi.org/10.1016/j.est.2019.100845.

15. D. Koppelhuber and J. Wall, "Ecology in Multi-Story Residential Buildings—Interdependency Between Ecology and Costs of Building Materials," in *10th International Structural Engineering and Construction Conference, ISEC 2019*, ISEC Press, Institute of Construction Management and Economics, Graz University of Technology, Graz, 2019, https://doi.org/10.14455/isec.res.2019.62.

16. M. V. Kljajić, A. S. Anđelković, V. Hasik, V. M. Munćan, and M. Bilec, "Shallow Geothermal Energy Integration in District Heating System: An Example from Serbia," *Renew Energy* 147 (2020): 2791–2800, https://doi.org/10.1016/j.renene.2018.11.103.

17. A. Ianchenko, K. Simonen, and C. Barnes, "Residential Building Lifespan and Community Turnover," *Journal of Architectural Engineering* 26, no. 3 (2020): 04020026, https://doi.org/10.1061/(ASCE)AE.1943-5568.0000401.

18. Damijan Miklavcic, ed., 8th European Medical and Biological Engineering Conference, EMBEC 2020*, November 29–December 3, 2020 Portorož, Slovenia: 80 (IFMBE Proceedings)*, vol. 80 (Springer, 2021).

19. J. Walzberg, T. Dandres, N. Merveille, M. Cheriet, and R. Samson, "Should We Fear the Rebound Effect in Smart Homes?," *Renewable and Sustainable Energy Reviews* 125 (2020): 109798, https://doi.org/10.1016/j.rser.2020.109798.

14708744, 2025, 5, Downloaded from https://iubmb.onlinelibrary.wiley.com/doi/10.1002/bab.2725 by University of Dundee, Wiley Online Library on [16/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

20. O. Jorgji, R. Di Bari, K. Lenz, J. Gantner, and R. Horn, "Analysing the Impact of Retrofitting and New Construction Through Probabilistic Life Cycle Assessment. A Method Applied to the Environmental-Economic Payoff Value of an Intervention Case in the Albanian Building Sector," in *Sustainable Built Environment D-A-CH Conference 2019: Transition Towards a Net Zero Carbon Built Environment, SBE 2019 Graz* (Institute of Physics Publishing, Institute for Acoustics and Building Physics, University of Stuttgart, 2019), https://doi.org/10.1088/1755-1315/323/1/012184.

21. P. C. Slorach and L. Stamford, "Net zero in the Heating Sector: Technological Options and Environmental Sustainability from Now to 2050," *Energy Conversation and Management* 230 (2021): 113838, https://doi.org/10.1016/j.enconman.2021.113838.

22. O. E. Gunaydin, E. Ertekin, and G. Gunaydin, "Four Weeks of Exercise Regimen for Sedentary Workers With Rounded Shoulder Posture: A Randomized Controlled Study," *Sao Paulo Medical Journal* 141, no. 3 (2023). https://doi.org/10.1590/1516-3180.2022.0257.r1.06072022.

23. R. K. R. Guduru, A. Domeika, and A. Domeikienė, "Effect of Rounded and Hunched Shoulder Postures on Myotonometric Measurements of Upper Body Muscles in Sedentary Workers," *Applied Sciences* 12, no. 7 (2022): 3333. https://doi.org/10.3390/app12073333.

24. R. K. R. Guduru, A. Domeika, M. Dubosiene, and K. Kazlauskiene, "Prediction Framework for Upper Body Sedentary Working Behaviour by Using Deep Learning and Machine Learning Techniques," *Soft Computing* 26, no. 23 (2021): 12969–12984. https://doi.org/10.1007/s00500-021-06156-8.

25. A. Weisman, and Y. Masharawi, "Does Altering Sitting Posture Have a Direct Effect on Clinical Shoulder Tests in Individuals with Shoulder Pain and Rotator Cuff Degenerative Tears?," *Physical Therapy*, 99, no. 2 (2018): 194–202. https://doi.org/10.1093/ptj/pzy111.

26. R. K. R. Guduru, A. Domeika, L. Obcarskas, and B. Ylaite, "The Ergonomic Association Between Shoulder, Neck/Head Disorders and Sedentary Activity: A Systematic Review," *Journal of Healthcare Engineering*, 2022 (2022): 1–15. https://doi.org/10.1155/2022/5178333.

27. A. G. Pascoal, A. Ribeiro, and J. Infante, "Scapular Resting Posture and Scapulohumeral Rhythm Adaptations in Volleyball Players: Implications for Clinical Shoulder Assessment in Athletes," *Sports* 11, no. 6 (2023): 114. https://doi.org/10.3390/sports11060114.

28. Y. Kobayashi, R. Matsuoka, T. Otokuni, T. Minami, and T. Watanabe, "Assessment of Reference Values for Shoulder Ultrasonography and Effects of Position Change on Shoulder Hemodynamics," *Journal of Clinical Ultrasound*. Portico (2025). https://doi.org/10.1002/jcu.23927.

29. A. Holtermann, V. Schellewald, S. E. Mathiassen, et al., "A Practical Guidance for Assessments of Sedentary Behavior at Work: A PEROSH Initiative," *Applied Ergonomics* 63(2017): 41–52. https://doi.org/10.1016/j.apergo.2017.03.012.

30. Y. Tian, D. Qiu, R. Song, et al., " Pathogenic Factor Analysis of Shoulder Periarthritis and Design of Virtual Reality Exercise Intervention System," *Applied Bionics and Biomechanics*, 2024, no. 1 (2024). Portico. https://doi.org/10.1155/abb/6543337.

31. R. S. Senthilkumar, B. Parthiban, and M. Parghavi, "Evaluation of Work Related Musculoskeletal Disorders in Shoulder and Neck With Ergonomic Intervention Among School Teachers," *Research Journal of Pharmacy and Technology*, 12, no. 8 (2019): 3726. https://doi.org/10.5958/0974-360x.2019.00637.1.

14708744, 2025, 5, Downloaded from https://iubmb.onlinelibrary.wiley.com/doi/10.1002/bab.2725 by University of Dundee, Wiley Online Library on [16/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License